DDCA PROJECT-1

DESIGN AND SIMULATION OF A DIGITAL DOOR ACCESS SYSTEM USING COMBINATIONAL LOGIC

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INTRODUCTION

- A digital door access system permits entry only when the entered code matches the authorized code.
- The system can be designed using combinational logic without memory elements.
- Main objective: design, simplify, and simulate the access-control logic.





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OBJECTIVES

- A digital door access system permits entry only when the entered code matches the authorized code.
- The system can be designed using combinational logic without memory elements.
- Main objective: design, simplify, and simulate the access-control logic.



SYSTEM BLOCK DIAGRAM

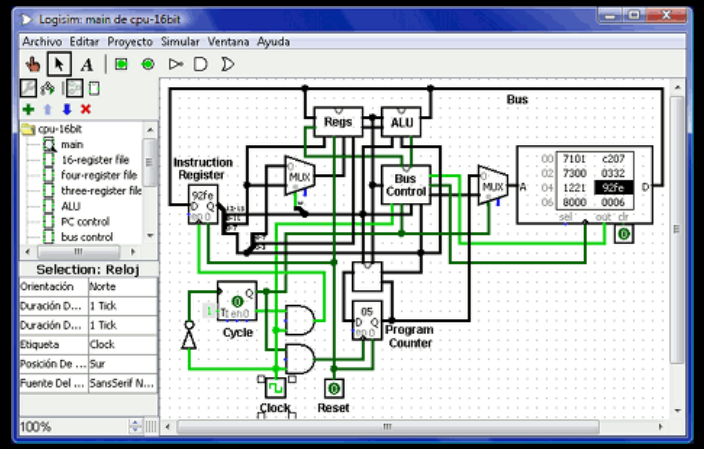
User / Keypad → Code Input Bits → Combinational Logic → VALID signal → Door Lock / LED

Optional: buzzer or red LED for an invalid code.

The combinational block produces its output immediately from the present inputs.



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DESIGN ASSUMPTION



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Use a 4-bit code input: A, B, C, D.

Authorized code selected for the example: 1011.

VALID = 1 only for 1011; otherwise, VALID = 0.

A practical system can expand this approach to more bits or multiple valid codes.

TRUTH TABLE

FOR EXAMPLE:

A B C D | VALID

0 0 0 0 | 0

0 0 0 1 | 0

0 0 1 0 | 0

0 0 1 1 | 0

0 1 0 0 | 0

0 1 0 1 | 0

0 1 1 0 | 0

0 1 1 1 | 0

1 0 0 0 | 0

1 0 0 1 | 0

1 0 1 0 | 0

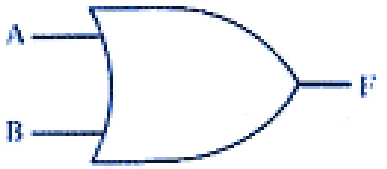
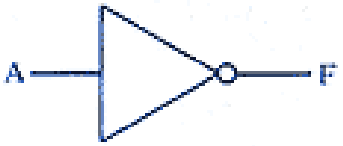
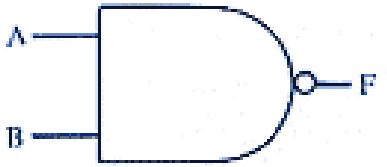
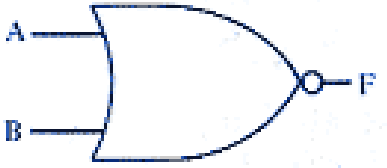
1 0 1 1 | 1

1 1 0 0 | 0

1 1 0 1 | 0

1 1 1 0 | 0

1 1 1 1 | 0

Name	Graphic Symbol	Algebraic Function	Truth Table
AND		F = A · B or F = AB	A B | F 0 0 | 0 0 1 | 0 1 0 | 0 1 1 | 1
OR		F = A + B	A B | F 0 0 | 0 0 1 | 1 1 0 | 1 1 1 | 1
NOT		$F = \overline{A}$ or F = A′	A | F 0 | 1 1 | 0
NAND		$F = (\overline{AB})$	A B | F 0 0 | 1 0 1 | 1 1 0 | 1 1 1 | 0
NOR		$F = (\overline{A + B})$	A B | F 0 0 | 1 0 1 | 0 1 0 | 0 1 1 | 0

❑ Boolean Expression:

For authorized code 1011:

VALID = A · B' · C · D

The complement B' is generated using a NOT gate.

Four conditions must be true simultaneously, so an AND gate generates VALID.

❑ Logic Circuit:

Inputs: A, B, C, D

B passes through a NOT gate to produce B'.

A, B', C, and D feed a 4-input AND gate.

AND output = VALID.

VALID can drive a green LED or a relay/driver stage for the lock

❑ Simulation Procedure:

Create the four digital input switches in the simulator.

Add one NOT gate and a 4-input AND gate.

Connect A, C, and D directly, and B through the inverter.

Monitor VALID with an LED or logic probe.

Test all 16 input combinations and confirm that only 1011 produces VALID = 1.

❑ Expected Simulation Results

Input 1011 ⟶ VALID = 1 ⟶ Door OPEN.

Input 1010 ⟶ VALID = 0 ⟶ Door LOCKED.

Input 0011 ⟶ VALID = 0 ⟶ Door LOCKED.

Input 1111 ⟶ VALID = 0 ⟶ Door LOCKED.

Thus, the circuit accepts only the programmed code.

ADVANTAGES AND LIMITATIONS

Advantages: simple, fast, low-cost, easy to simulate and implement.

Limitation: a fixed combinational code is not suitable for high-security applications.

No memory means there is no attempt counter, code storage, or event logging.

For real systems, use sequential logic or a microcontroller with secure authentication.

Applications

Laboratory and classroom access-control demonstrations.

Cabinet, locker, and small enclosure access.

Digital logic training and FPGA/HDL demonstrations.

Basic keypad-controlled electronic locks.

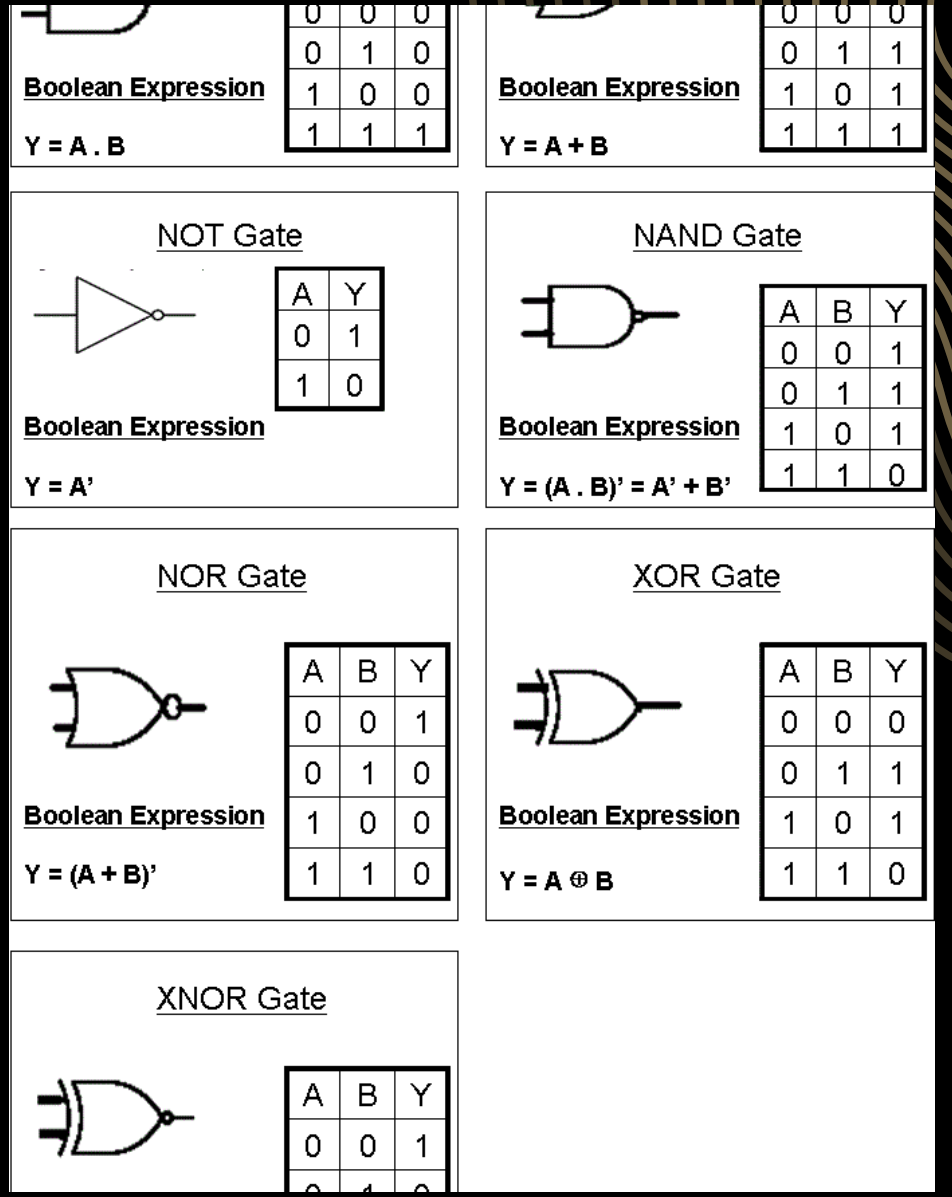
CONCLUSION

A DIGITAL DOOR ACCESS SYSTEM CAN BE IMPLEMENTED USING A SIMPLE COMBINATIONAL CIRCUIT.

FOR THE EXAMPLE CODE 1011, VALID = A·B'·C·D.

SIMULATION VERIFIES THAT ONLY THE AUTHORIZED COMBINATION ACTIVATES THE DOOR-OPEN OUTPUT.

THE DESIGN DEMONSTRATES PRACTICAL USE OF BOOLEAN ALGEBRA AND LOGIC GATES.



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THANK YOU